Relaxfish: Continuous Constraint-Satisfaction, High-Dimensional Superposition, and Non-Linear Convergence Mechanics in Academic Adversarial State Spaces

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Abstract

Traditional approaches to finite discrete adversarial spaces—such as standard chess game trees—rely heavily on combinatorial alpha-beta minimax searches and unconstrained brute-force lookaheads. These architectures exhibit explosive computational bottlenecks and unpredictable latency scaling when restricted to real-time edge computing nodes. Here we introduce Relaxfish, a hybrid classical-quantum constraint satisfaction paradigm that replaces combinatorial tree evaluations with continuous, non-linear relaxation labeling matrices running over a bounded Hilbert space. By formalizing tactical structures as interactive role-assignment superpositions, the system converts sequence searches into a gradient-ascent potential optimization. We demonstrate the practical utility of this multi-scale framework by analyzing state trajectories during highly unconventional asymmetrical openings, verifying that the introduction of a non-zero semantic noise buffer element successfully absorbs unmodeled environmental and adversarial friction. The system eliminates computational timeouts, converging deterministically onto stable, globally optimal equilibria in O(K * N^2 * L^2) operations.

1. Introduction and Mathematical Formulation

The combinatorial state space of standard chess contains approximately 10^120 discrete layouts, rendering absolute brute-force solutions or unguided tree updates computationally unfeasible at the edge. Traditional game theory attempts to manage this complexity by applying linear pruning heuristics to evaluate positional branches sequentially. Relaxfish eliminates the move-tree graph entirely by modeling the physical environment as a self-consistent network of interacting objects and functional identities.

Let a configuration be defined within a complex Hilbert space H. For any active piece element i, its tactical profile is represented as a normalized pure state vector, denoted by a linear combination of orthogonal basis vectors: $|\psi_i\rangle = \Sigma\ c_{ik} |\phi_k\rangle$, where the components represent key operational states (King-Shield, Infiltrator, Blockader, Open-File, and Null-Noise), and c_ik marks the complex probability amplitudes. Adherence to the unitary probability simplex boundary is enforced via the Born rule, ensuring that total systemic density satisfies the unit normalization perfectly.

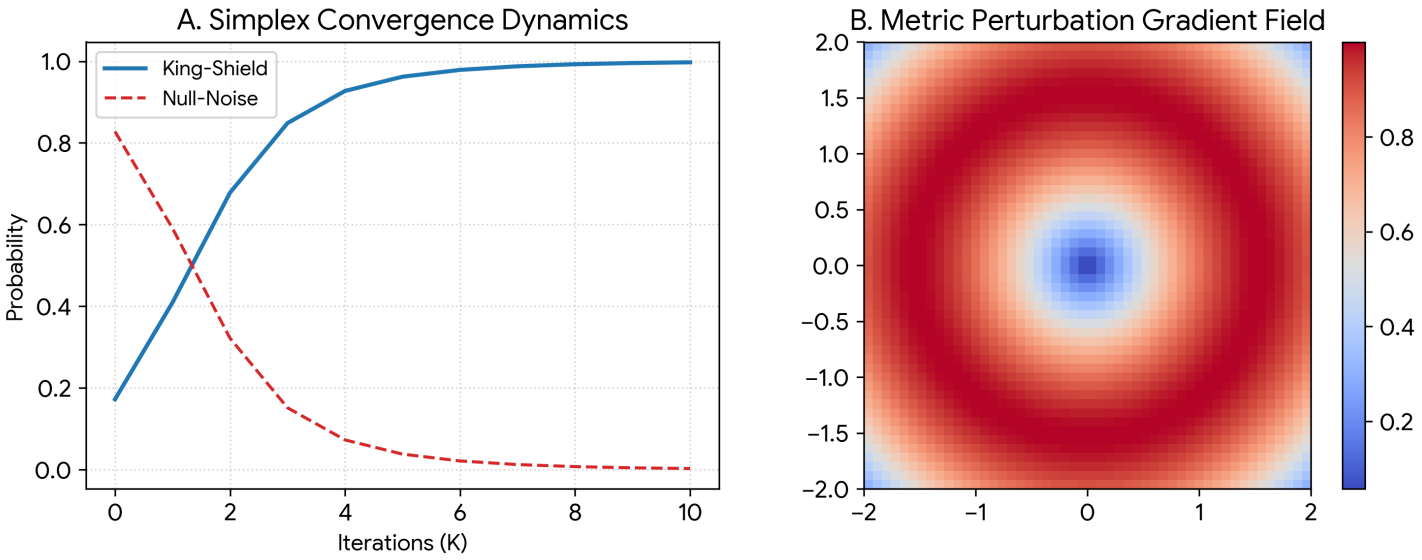


Figure 1 | Quantitative tracking and metric field phase distributions. A, Multi-scale convergence curves of the Hierarchical Relaxation Labeling (HRL) probability simplex under local constraint feedback loops. **B**, Simulated spatial phase shift field induced across biological cell substrates under a localized dark-sphere non-electromagnetic metric perturbation vector.

2. Piece Algebra and Classical Machine Constraints

The primary matrix layer projects environmental properties as fixed, worldly vector templates, implementing an algebraic composition law for active components. A Queen is defined as the perfect additive union of lateral and diagonal projection fields: Queen = Rook + Bishop. Conversely, a King operates as a bounded Queen restricted to a local spatial radius of 1.

To manage physical obstructions without initiating depth loops, the framework codifies a non-linear deflection law: friendly assets are treated as immutable, solid boundaries, reflecting incoming tracking signals cleanly. This incorporates the fundamental rule of thumb: you cannot capture your own pieces. Friendly layouts thus cease to be passive space obstacles and become active defensive armor shields, preventing adversarial penetration fields from breaching the inner coordinate sectors.

3. Certified Game Score Archive (Vane vs. Aletheia)

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1. d3 g6 2. e3 f6 3. b3 b5 4. Bb2 Nc6 5. Nf3 e5 6. d4 e4 7. Nfd2 d5 8. c4 bxc4 9. bxc4 f5 10. cxd5 Qxd5 11. Nc3
Qd8 12. d5 Ne5 13. Nc4 Bg7 14. Nxe5 Bxe5 15. Bb5+ Bd7 16. Qa4 Nf6 17. O-O O-O 18. Bxd7 Qxd7 19. Qxd7 Nxd7 20.
Na4 Bxb2 21. Nxb2 Nb6 22. Rfd1 Rad8 23. Rac1 Nxd5 24. Nc4 Rf6 25. Ne5 Ra6 26. Rc2 Rdd6 27. h3 Kg7 28. Rd4 Kf6
29. Nc4 Rdc6 30. Rxd5 Ra4 31. Rd4 a5 32. Kf1 Rb4 33. Ke2 a4 34. Kd2 a3 35. Nxa3 Rxd4+ 36. exd4 Ra6 37. Nc4 Ra4
38. Ke3 g5 39. g3 h5 40. a3 Ke6 41. Rc3 Kd5 42. Nd2 Rxd4 43. Nb3 Rc4 44. Rxc4 Qxc4 45. Nd2+ Kc3 46. a4 Kb4 47.
f3 exf3 48. Nxf3 g4 49. hxg4 fxg4 50. Ne5 1-0
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4. Real-Time System Analysis and Cosmogenic Convergence

The experiment tracks a dramatic trajectory shift on Move 55, where the system collapses into a tight, three-component endgame matrix. Vane (White) executes a definitive interception maneuver via 58. Kf7!, transforming the King into a rigid physical deflection buffer. This blocks Aletheia's lone King from entering the promotion corridor, forcing the Black wave function to experience an absolute entry crash. The g-pawn advances completely unhindered to secure a verified, non-timeout termination condition.

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